

Playwright Test Lead Training Program

2-Month Comprehensive Mastery Program (GenAI-Native)

Program Mentor

Sudhakar Kakunuri (QTP Sudhakar)

Product QA Head | India's First Playwright Corporate Trainer | 20+ Years Experience | 10,000+ Engineers Trained | GenAI & Agentic Testing Pioneer

Known globally as "QTP Sudhakar", Sudhakar Kakunuri teaches "What is Needed" for the role, not just "What is there" in the tool. He focuses on hand-holding students to achieve lead-level expertise.

Next Batch: January 19, 2026	Duration: 2 Months (60 Continuous Sessions)	Schedule: Monday to Friday (1.5 hours/day)
Format: Live Interactive + Hands-on Labs + Weekend Practice	Batch Size: Limited to 15 seats for personal attention	Mode: Online

Program Philosophy

Moving beyond generic tutorials. This program is designed to bridge the "Mentorship Gap" by providing Level 3 Hand-holding, focusing on the **Mindset** and **Practical Application** required for Test Lead roles.

The Test Lead Advantage

- Role-Specific:** Learn what's needed for Lead roles, not just junior engineering tasks.
- Practice-First:** Daily in-class practice labs (Warm-up, Core, and Open Challenges).
- AI-Native:** Real workflows with Cursor, Copilot, and AI-powered self-healing.
- Vibe Testing:** Master proprietary exploratory testing methodologies.
- Career Support:** Mock interviews, resume strategy, and listing on VibeTestQ.com.

Course Curriculum (60 Sessions)

The AI-First Test Lead Mindset

Program Inauguration & The Evolution of Testing

- The Evolution of Testing in the GenAI Era
- Why Playwright? Comparative analysis with legacy tools
- The Mentorship Gap: Why self-learning via YouTube often fails
- "What is Needed" vs "What is There" mindset

Warm-up Lab: Assessing current skills and setting up the Lead-level environment

Playwright Architecture & Core Fundamentals

- Playwright architecture: Under the hood
- Browser-Context-Page hierarchy and test isolation
- Context-based execution vs Browser-based execution
- Page navigation, URL handling, and load states

Core Practice: multi-context configuration for enterprise isolation

Git Basics & Node.js Project Structure

- Git for beginners: Init, Config, Add, Commit, Push, Pull
- Node.js runtime & NPM ecosystem for Test Leads
- Package management, semantic versioning, and scripts
- Project organization and secure config management

Core Practice: Building a version-controlled NPM automation toolbox

Modern JS/TS for Enterprise Scale - Part 1

- Essential JS for Playwright (Async/Await, ES6+ features)
- Scope management and Module patterns
- TypeScript Fundamentals: Types, Interfaces, and Enums

Open Challenge: Async data orchestration scripts with TS

Modern JS/TS for Enterprise Scale - Part 2

- Type safety in locators, configurations, and utilities
- Building reusable business logic helpers with TS
- Data manipulation for complex test scenarios

Core Practice: Converting legacy JS logic to Type-safe TS functions

Playwright Mastery & Engineering Excellence

Development Environment & Advanced Locators

- VS Code setup: Extensions and professional configuration
- Locators Mastery: Role-based, Accessibility, and Resilient CSS/XPath
- Codegen: Recording tests and generating scaffolding
- Handling dynamic and non-deterministic elements

Warm-up: Auditing applications for locator resilience and accessibility

Actionability, Waits & Assertions

- The Playwright actionability check and auto-waiting
- Web-first assertions: Retrying expectations for stability
- Custom waiting strategies for complex SPAs
- Actionability, timeouts, and non-flaky execution

Core Practice: Solving "flaky" UI scenarios with Playwright secrets

Mouse, Keyboard & File Operations

- Mastering user interactions: Hover, drag-and-drop, shortcuts
- Keyboard actions and complex input simulations
- File operations: Resilient uploads and downloads

Open Challenge: Interacting with complex third-party widgets and file systems

Multi-Context, Secure State & Auth

- Browser Contexts for parallel isolated testing
- Authentication state sharing (StorageState)
- Parallelism at scale vs sequential execution patterns

Core Practice: Building a multi-user login speed-optimized suite

UI Mode & Professional Debugging

- Playwright UI Mode: Time-travel debugging and interactive testing
- Inspector, Trace Viewer, and Log analysis
- Event handling and observability in test execution

Open Challenge: Debugging a broken multi-context workflow using Trace Viewer

Enterprise Framework Architecture

Complex UI: Frames, Dialogs & Windows

- Shadow DOM navigation and automation
- Handling nested iframes and multi-tab workflows
- Dialogs, Popups, and Window management

Core Practice: Refactoring spaghetti code into Component-based POM

Playwright Fixtures & Dependency Injection

- Understanding Fixtures and dependency injection
- Custom Fixtures for application-specific state
- Extending Playwright Test with custom logic

Open Challenge: Implementing a 'Logged-in-User' fixture for all tests

Configuration Mastery & Project Matrices

- Advanced Playwright.config patterns
- Multi-environment execution (Dev, QA, Staging, Prod)
- Configuration for different viewports and devices

Core Practice: Setup a multi-project, multi-environment config system

Enterprise Test Data Strategies

- Managing test data via JSON, CSV, and Excel
- Dynamic data generation and Faker integration
- Parameterization at the project level

Core Practice: Build a dynamic data-driven suite for bulk data entry

Visual Regression & Accessibility Automation

- Visual regression testing with pixel/AI comparisons
- Accessibility validation (axe-core) for compliance
- Implementing visual testing for responsive design

Open Challenge: Designing a 'Lead Dashboard' style report with Visual & Auth artifacts

The AI-Native Automation Engineer

GenAI Foundations & Prompt Engineering for QA

- Prompt Engineering for Test Case generation
- Context Engineering: Providing app knowledge to LLMs
- Avoid AI Hallucinations in Test Logic
- LLMs and the current state of AI in testing

Warm-up: Generating test scenarios from raw requirements using AI

AI IDEs Mastery: Cursor, Copilot & AntiGravity

- Cursor for Framework development and chat logic
- Copilot Chat for complex code generation
- Integrating AI directly into the Lead IDE workflow
- AI-first development environments

Core Practice: Using LLMs to build 10 complex tests in 1 hour

AI-Powered Test Artifact Generation

- Gherkin/Manual steps to Playwright TS conversion
- Building AI-assisted test skeletons and POMs
- Reviewing and validating AI-generated code

Open Challenge: Building a simple 'Self-Healing' helper with AI

Testing AI Applications & Workflows

- Testing LLM applications, RAG systems, and Chatbots
- Non-deterministic UI validation and hallucination testing
- Specialized frameworks for AI product testing

Core Practice: Building a test suite for an AI-powered portal

Mid-Program Assessment: Scalable POM + AI

- Project architecture review and feedback
- Lead-level code reviews and maintenance best practices
- Enterprise-grade patterns check

Capstone: Build a 5-page enterprise site framework with AI-assisted utilities

End-to-End API & Integration Testing

API Testing Mastery with Playwright

- REST/GraphQL testing using APIRequestContext
- JSON Schema validation and response assertions
- Managing API state, tokens, and headers

Core Practice: Building a robust API test suite from scratch

Hybrid Integration: UI + API Workflows

- Speeding up UI tests with API preconditions
- Bypassing Login via API for UI automation
- Data verification and cleanup using API calls

Open Challenge: Automate a UI flow where data setup is 100% API-driven

AI Planner Agent: Intelligent Test Strategy

- Introduction to Playwright AI Agents
- Planner Agent: Automated test planning and coverage analysis
- Intelligent test strategy generation from requirements

Core Practice: Generating a comprehensive test strategy using Planner Agent

AI Generator & Healer Agents

- Generator Agent: Automated test script & POM creation
- Healer Agent: Self-healing maintenance and locator repair
- Integrating agents into the development workflow

Core Practice: Building and repairing a complex suite using specialized AI Agents

Script Evaluation & Page Context Mastery

- Mastering `page.evaluate()` and `page.evaluateHandle()`
- Executing JavaScript in browser context for advanced checks
- Accessing `JSHandle` and DOM elements via code

Challenge: Automate complex dynamic behaviors via evaluation injection

Advanced Specialized Testing & AI Strategy

Vibe Testing: The Exploratory Methodology

- Session-based testing vs Scripted testing
- Creating Test Charters for the modern Lead
- AI-augmented exploration: Using AI as a testing partner

Workshop: Conducting a 'Vibe Testing' session on a new product

Framework Engineering: Parallelism & Sharding

- Optimizing execution speed via projects and sharding
- Reporting Strategy: Allure, custom reporters, and CI artifacts
- Test annotations, tagging, and execution control

Core Practice: Setup a multi-sharded parallel CI pipeline

Maintenance & Best Practices for Scale

- Scalable framework architecture and structure
- Test maintenance strategies and best practices
- Code review checklists for automation squads

Core Practice: Multi-resolution Visual Regression suite setup

Cloud Execution: LambdaTest & BrowserStack

- Executing on real devices and cloud grids
- Interpreting cloud analytics and results at scale
- Cost vs Benefit of Cloud infrastructure

Open Challenge: Running parallel tests on LambdaTest/BrowserStack grid

Agent Orchestration & Model Context Protocol (MCP)

- Orchestrating multiple Agents (Planner, Generator, Healer)
- Model Context Protocol (MCP) for deep AI integration
- AI Governance and responsibility in testing

Workshop: Designing an 'AI Testing Strategy' with MCP integration

DevOps, CI/CD & Pipeline Engineering

Advanced Git & Team Lead Collaboration

- Advanced Branching strategies (Gitflow, Trunk-based)
- Collaborative PR workflows and Lead-level code reviews
- Conflict resolution and version control best practices

Core Practice: Simulating a complex multi-lead PR review and merge lifecycle

GitHub Actions & Pipeline Engineering

- YAML configuration for parallel Playwright execution
- CI/CD integration, scheduling, and failure management
- Automated artifact uploads and reporting triggers

Core Practice: Building a delivery pipeline from scratch

Docker & Containerization for QA

- Creating custom Playwright Docker images
- Running tests in isolated, reproducible containers
- Integration with enterprise CI tools

Open Challenge: Build and execute a suite inside a local Docker container

Jenkins & Enterprise CI Visibility

- Jenkins pipelines (Groovy Scripting) and security
- Integrating with Azure DevOps or GitLab pipelines
- Secret management and secure pipeline design

Workshop: Connecting Playwright to an enterprise Jenkins instance

Network Interception & API Mocking Deep Dive

- Mastering request routing and response mocking
- Simulating performance bottlenecks and server errors
- API simulation for isolated UI testing

Core Practice: Mocking a 3rd party gateway for reliable tests

The Lead Persona & Career Transformation

Capstone Project: Enterprise Architecture

- Requirement analysis and tool selection
- Designing a scalable, AI-ready framework skeleton
- Applying all enterprise patterns learned

Development: Framework skeleton and architecture setup

Capstone Project: Implementation Intensity

- End-to-end implementation with AI-native workflows
- Integrating agents, fixtures, and dynamic reporting

Development: Complete suite implementation with 1:1 mentor feedback

Resume Strategy & LinkedIn Branding

- Crafting a 'Test Lead' resume highlighting AI skills
- LinkedIn profile optimization and portfolio showcase
- Positioning for success in AI-augmented roles

Workshop: 1:1 Resume review session

Mock Interviews & Salary Negotiation

- Common Test Lead and AI-automation interview questions
- Experience gap handling and salary negotiation techniques

Lab: Live mock interview session with feedback

Graduation & Career Launch

- Final project showcase and certification ceremony
- Listing on VibeTestQ.com verified practitioners

Event: Program graduation and next steps guidance

Learning Methodology

The Support System

- **Daily Labs:** Dedicated 1.5 hours of guided practice every day.
- **Mentor Hand-holding:** Direct access to Sudhakar for code reviews and logic guidance.
- **Collaborative Sessions:** See how peers solve the same problems to broaden your perspective.
- **Lifetime Community:** Join an elite group of Playwright Test Leads for ongoing industry trends.

Total Learning Outcomes

By the end of 60 sessions, participants will have:

- ✓ Built production-ready Playwright framework
- ✓ Integrated AI-powered testing capabilities
- ✓ Implemented CI/CD pipelines
- ✓ Created comprehensive test suites
- ✓ Developed professional portfolio
- ✓ Gained industry-ready automation skills
- ✓ Network with industry professionals
- ✓ Learned from top industry experts

Playwright Test Lead Training Program

For enrollment inquiries, please send whatsapp message to +91-8328152362